

Supplementary Material

Climate-Signal Preservation of Bias-Corrected Daily Precipitation from Global Climate Models: A Comparison of Quantile Mapping Methods over Uttaradit, Thailand

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Supplementary tables are numbered as they are cited in the manuscript. Where a listing runs to thousands of rows it is supplied as a machine-readable file with this submission rather than as a printed table; those files are listed at the end.

Supplementary Table S1. Design of the three bias-correction methods and the settings they share.

Item	Description
Method: QM/EQM	core purpose: Historical bias correction; distribution treatment: Empirical marginal quantile mapping onto the observed calibration distribution; climate-signal property: Not explicitly preserved; temporal reconstruction: No
Method: DetQM	core purpose: Bias correction with mean-signal preservation; distribution treatment: Target-period wet-day mean removed multiplicatively, quantile-mapped, then restored; climate-signal property: Mean-signal oriented; temporal reconstruction: No
Method: QDM	core purpose: Bias correction with quantile-signal preservation; distribution treatment: Observed calibration quantile multiplied by the model's own quantile-level change ratio; climate-signal property: Quantile-signal oriented; temporal reconstruction: No
Wet-day threshold	1.0 mm d ⁻¹ , identical for all methods
Zero-rain treatment	days at or below the frequency-adapted model threshold are set to zero; observed zeros are never imputed
Frequency adaptation	model wet-day threshold raised until the model wet-day frequency over the calibration period matches the observed frequency, fitted separately for each gauge-model pair
Empirical CDF	non-parametric, sorted wet-day sample
Plotting positions	hazen, $(i - 0.5)/n$
Interpolation	linear within the calibration quantile range
Extrapolation	QM and DetQM hold the multiplicative correction ratio of the nearest calibration quantile; QDM applies the model's own change ratio, which extends naturally beyond the range. No value is clipped.
Calibration period	1981–2002, parameters frozen thereafter
Independent validation	2003–2014, no refitting
Future baseline	1995–2014, same model and same method as the future series
Future period	2021–2050, recomputed from the daily corrected series
Minimum wet days to fit	30
Mean wet days extrapolated	0.41% (maximum 10.6% for one gauge-model pair)

Supplementary Table S2. Calibration-period mean absolute bias by index and method, averaged over 13 gauges and 7 GCMs. Station- and model-level values for all 16 indices are in Supplementary Data S2.

Index	Raw CMIP6	QM/EQM	DetQM	QDM
Wet days	71.14	0.03	0.46	0.03
PRCPTOT	40.82	1.19	1.2	1.19
SDII	45.25	0.01	0.42	0
q50	46.92	0.02	0.63	0.02
q90	49.93	0.01	0.23	0.01
q95	48.17	0.02	0.19	0.02
q99	51.57	0.08	0.15	0.08
Rx1day	49.11	7.27	7.27	7.27
Rx5day	34.52	25.52	25.52	25.52
R10mm	58.26	1.19	1.24	1.19
R20mm	74.42	1.38	1.46	1.36
R50mm	194.93	1.04	1.05	1.04
R95p	74.96	1.19	1.33	1.19
R99p	95.37	1.35	1.35	1.35
CDD	47.9	42.52	42.21	42.52
CWD	89.99	35.05	35.06	35.05

Supplementary Table S3. Independent-validation mean absolute bias by index and method, using parameters fitted on the calibration period and frozen. Station- and model-level values are in Supplementary Data S3.

Index	Raw CMIP6	QM/EQM	DetQM	QDM
Wet days	67.22	28.65	28.76	28.65
PRCPTOT	38.45	15.9	15.1	15.82
SDII	44.08	26.91	26.97	27.37
q50	43.2	24.17	26.31	26.52
q90	47.24	32.31	32.86	33.36
q95	47.95	31.17	31.93	33.69
q99	54.81	28.8	27.39	27.61
Rx1day	47.38	30.96	29.98	26.85
Rx5day	35.63	33.59	33.08	31.52
R10mm	53.39	17.79	17.47	18.75
R20mm	70.25	35.95	36.12	37.27
R50mm	124.84	60.71	61.48	61.46
R95p	74.16	41.81	43.51	45.48
R99p	88.64	58.74	60.59	60.76
CDD	48.42	51.78	51.98	51.47
CWD	83.19	38.4	38.48	38.73

Supplementary Table S4. Climate-signal preservation summary for every scenario, index and method.

Scenario	Index	Method	n Gauges	Median PE (%)	IQR (%)	Mean (%)	SD (%)
SSP2-4.5	Wet days	QM/EQM	7 13	0.29	-2.02 to 2.22	0.06	3.27
SSP2-4.5	Wet days	DetQM	7 13	0.84	-1.77 to 2.35	0.41	3.59
SSP2-4.5	Wet days	QDM	7 13	-0.09	-2.18 to 2.50	-0.04	3.35
SSP2-4.5	PRCPTOT	QM/EQM	7 13	1.26	0.30 to 3.78	1.5	2.62
SSP2-4.5	PRCPTOT	DetQM	7 13	0.81	-0.23 to 2.25	0.47	2.41
SSP2-4.5	PRCPTOT	QDM	7 13	0.73	-0.18 to 2.12	0.97	1.74
SSP2-4.5	SDII	QM/EQM	7 13	0.94	0.32 to 2.77	2	4.94
SSP2-4.5	SDII	DetQM	7 13	-0.09	-1.34 to 1.64	0.61	4.94
SSP2-4.5	SDII	QDM	7 13	0.69	-1.04 to 2.13	1.57	4.63
SSP2-4.5	q50	QM/EQM	7 13	4.18	0.72 to 7.36	3.96	5.04
SSP2-4.5	q50	DetQM	7 13	0.88	-1.36 to 5.78	1.32	6.4
SSP2-4.5	q50	QDM	7 13	1.22	0.89 to 6.34	3.25	4.58
SSP2-4.5	q90	QM/EQM	7 13	0.16	-0.12 to 3.48	3.12	6.47

Scenario	Index	Method	n Gauges	Median PE (%)	IQR (%)	Mean (%)	SD (%)
SSP2-4.5	q90	DetQM	7 13	1.03	-0.81 to 3.34	2.53	5.64
SSP2-4.5	q90	QDM	7 13	0.43	-0.78 to 3.63	2.84	6.26
SSP2-4.5	q95	QM/EQM	7 13	0.09	-1.80 to 2.03	1.17	4.16
SSP2-4.5	q95	DetQM	7 13	-0.02	-1.43 to 1.63	0.74	2.98
SSP2-4.5	q95	QDM	7 13	-0.05	-1.03 to 1.29	0.78	2.92
SSP2-4.5	q99	QM/EQM	7 13	1.96	-0.83 to 2.93	0.73	11.27
SSP2-4.5	q99	DetQM	7 13	1.46	-1.29 to 2.11	0.17	9.73
SSP2-4.5	q99	QDM	7 13	0.26	-2.21 to 2.08	0.26	8.32
SSP2-4.5	Rx1day	QM/EQM	7 13	1.63	0.34 to 1.97	-0.86	6.29
SSP2-4.5	Rx1day	DetQM	7 13	0.81	-0.57 to 1.21	-1.24	5.09
SSP2-4.5	Rx1day	QDM	7 13	0.59	-0.47 to 1.13	-0.06	2.69
SSP2-4.5	R10mm	QM/EQM	7 13	-0.8	-2.34 to 4.77	1.22	5.54
SSP2-4.5	R10mm	DetQM	7 13	-2.44	-3.05 to 1.56	-0.56	5
SSP2-4.5	R10mm	QDM	7 13	-2.03	-2.42 to 1.62	-0.05	4.34
SSP2-4.5	R20mm	QM/EQM	7 13	1	-1.72 to 4.87	1.17	14.28
SSP2-4.5	R20mm	DetQM	7 13	0.52	0.14 to 3.12	1.49	13.58
SSP2-4.5	R20mm	QDM	7 13	2.11	-0.77 to 6.09	3.75	9.26
SSP2-4.5	R50mm	QM/EQM	6 9–12	15.54	-4.83 to 25.34	16.18	57.65
SSP2-4.5	R50mm	DetQM	6 9–12	6.6	1.90 to 23.87	12.32	50.94
SSP2-4.5	R50mm	QDM	6 9–12	14.76	10.55 to 18.52	14.98	48.03
SSP2-4.5	R95p	QM/EQM	7 6–13	-1.49	-3.43 to 2.32	1.55	29.97
SSP2-4.5	R95p	DetQM	7 6–13	-2.59	-4.10 to 0.81	-0.56	24.85
SSP2-4.5	R95p	QDM	7 6–13	-1.45	-3.95 to 0.08	1.07	27.97
SSP2-4.5	R99p	QM/EQM	7 5–13	-9.21	-19.36 to 3.38	-6.33	18.99
SSP2-4.5	R99p	DetQM	7 5–13	-5.76	-9.09 to -0.23	-3.35	20.5
SSP2-4.5	R99p	QDM	7 5–13	-5.95	-14.19 to 18.91	5.45	34.4
SSP2-4.5	Rx5day	QM/EQM	7 13	0.83	0.31 to 1.71	-0.12	6.71
SSP2-4.5	Rx5day	DetQM	7 13	1.09	-1.42 to 1.38	-0.71	5.56
SSP2-4.5	Rx5day	QDM	7 13	-0.15	-1.51 to 0.66	-0.18	4.47
SSP2-4.5	CDD	QM/EQM	7 13	-2.41	-5.74 to 0.30	-1.38	5.78
SSP2-4.5	CDD	DetQM	7 13	-3.1	-6.24 to 0.69	-1.68	6.18
SSP2-4.5	CDD	QDM	7 13	-3.32	-5.44 to 1.26	-1.07	6.22
SSP2-4.5	CWD	QM/EQM	7 13	2.8	-1.97 to 6.70	2.6	6.25
SSP2-4.5	CWD	DetQM	7 13	2.79	-2.11 to 6.71	2.9	6.53
SSP2-4.5	CWD	QDM	7 13	1.73	-1.89 to 7.08	2.79	6.62
SSP5-8.5	Wet days	QM/EQM	7 13	-0.97	-1.75 to -0.11	-0.79	1.43
SSP5-8.5	Wet days	DetQM	7 13	-1.25	-1.62 to -0.07	-0.73	1.49
SSP5-8.5	Wet days	QDM	7 13	-1.48	-1.74 to -0.11	-0.74	1.59
SSP5-8.5	PRCPTOT	QM/EQM	7 13	1.01	-0.23 to 2.05	2.17	4.14
SSP5-8.5	PRCPTOT	DetQM	7 13	1.19	-0.04 to 1.72	1.73	2.85
SSP5-8.5	PRCPTOT	QDM	7 13	-0.07	-0.36 to 0.64	1.2	3.27
SSP5-8.5	SDII	QM/EQM	7 13	1.46	0.08 to 3.06	3.07	5.93
SSP5-8.5	SDII	DetQM	7 13	1.17	0.22 to 2.17	2.52	4.32
SSP5-8.5	SDII	QDM	7 13	1.12	0.09 to 1.86	2.08	4.78
SSP5-8.5	q50	QM/EQM	7 13	2.33	2.16 to 3.79	3.38	3.13
SSP5-8.5	q50	DetQM	7 13	1.88	0.81 to 4.05	2.62	3.17
SSP5-8.5	q50	QDM	7 13	1.84	0.43 to 3.63	2.76	3.25
SSP5-8.5	q90	QM/EQM	7 13	1.21	-2.05 to 4.03	4.37	11.66
SSP5-8.5	q90	DetQM	7 13	1.46	-0.80 to 4.21	4.01	8.76
SSP5-8.5	q90	QDM	7 13	1.7	-0.72 to 3.97	4.15	9.42
SSP5-8.5	q95	QM/EQM	7 13	2.72	1.79 to 5.13	3.32	4.28
SSP5-8.5	q95	DetQM	7 13	2.78	1.72 to 3.17	2.2	2.46
SSP5-8.5	q95	QDM	7 13	0.76	0.51 to 2.69	1.26	2.39
SSP5-8.5	q99	QM/EQM	7 13	3.6	-1.32 to 7.53	4.42	7.41
SSP5-8.5	q99	DetQM	7 13	3.71	-1.19 to 6.64	3.59	5.75
SSP5-8.5	q99	QDM	7 13	0.04	-1.72 to 2.99	0.83	3.61
SSP5-8.5	Rx1day	QM/EQM	7 13	0.82	0.64 to 3.13	1.98	2.08

Scenario	Index	Method	n Gauges	Median PE (%)	IQR (%)	Mean (%)	SD (%)
SSP5-8.5	Rx1day	DetQM	7 13	1.34	0.58 to 3.15	2.15	2.17
SSP5-8.5	Rx1day	QDM	7 13	0.03	-0.71 to 1.10	0.16	1.78
SSP5-8.5	R10mm	QM/EQM	7 13	3.71	-0.56 to 6.88	5.17	10.74
SSP5-8.5	R10mm	DetQM	7 13	4.26	-1.42 to 8.18	4.62	7.57
SSP5-8.5	R10mm	QDM	7 13	0.48	-0.04 to 6.51	4.56	8.02
SSP5-8.5	R20mm	QM/EQM	7 13	6.53	4.40 to 10.37	6.72	5.86
SSP5-8.5	R20mm	DetQM	7 13	5.24	2.22 to 7.93	5.46	5.88
SSP5-8.5	R20mm	QDM	7 13	5.73	-1.03 to 9.46	4.28	7.25
SSP5-8.5	R50mm	QM/EQM	6 9–12	5.29	4.03 to 13.12	14.59	31.34
SSP5-8.5	R50mm	DetQM	6 9–12	9.67	-0.79 to 20.26	11.69	22.67
SSP5-8.5	R50mm	QDM	6 9–12	8.38	-10.86 to 18.38	5.19	19.62
SSP5-8.5	R95p	QM/EQM	7 6–13	-2.85	-6.95 to 6.04	2.9	17.08
SSP5-8.5	R95p	DetQM	7 6–13	-1.74	-4.06 to 6.80	1.54	9.31
SSP5-8.5	R95p	QDM	7 6–13	-6.21	-8.23 to -3.05	-6.49	5.47
SSP5-8.5	R99p	QM/EQM	7 5–13	13.86	-18.45 to 17.46	4.84	28.25
SSP5-8.5	R99p	DetQM	7 5–13	4.75	-14.57 to 24.42	5.14	29.75
SSP5-8.5	R99p	QDM	7 5–13	-5.98	-15.71 to 21.18	5.18	34.97
SSP5-8.5	Rx5day	QM/EQM	7 13	1.02	-0.19 to 4.74	2.22	3.14
SSP5-8.5	Rx5day	DetQM	7 13	1.36	-0.42 to 3.17	1.8	2.66
SSP5-8.5	Rx5day	QDM	7 13	-0.3	-0.65 to 1.35	0.22	1.42
SSP5-8.5	CDD	QM/EQM	7 13	2.58	-3.68 to 7.09	1.05	9.21
SSP5-8.5	CDD	DetQM	7 13	2.61	-3.29 to 6.82	1.04	9.34
SSP5-8.5	CDD	QDM	7 13	2.25	-3.03 to 6.77	0.92	9.2
SSP5-8.5	CWD	QM/EQM	7 13	-1.97	-4.50 to 3.64	0.25	9.08
SSP5-8.5	CWD	DetQM	7 13	-1.61	-4.60 to 3.79	0.36	9.38
SSP5-8.5	CWD	QDM	7 13	-1.56	-4.17 to 3.52	0.58	9.47

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23 **Supplementary Table S5.** Climate-signal preservation for every model, gauge, scenario and index. This listing has 9,804
24 rows and is supplied as a machine-readable file with this submission rather than as a printed table.

Contents	Rows	Columns	File
signal-preservation error per model, gauge, scenario and index	9804	11	Supplementary_Data_S5.csv

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26 **Supplementary Table S6.** Descriptive two-factor sum-of-squares partition for all indices and both scenarios.

Scenario	Index	N cells	GCMs	GCM main effect (%)	BC-method main effect (%)	Residual (%)
SSP2-4.5	CDD	273	7	49.9	0.1	50
SSP2-4.5	CWD	273	7	52.9	0	47
SSP2-4.5	PRCPTOT	273	7	78.3	0.6	21.1
SSP2-4.5	PRCPTOT_dry	273	7	77.4	0.1	22.6
SSP2-4.5	PRCPTOT_wet	273	7	67.8	0.6	31.6
SSP2-4.5	R10mm	273	7	73.5	0.9	25.6
SSP2-4.5	R20mm	273	7	9.5	0.3	90.2
SSP2-4.5	R50mm	214	7	18.5	0.2	81.3
SSP2-4.5	R95p	273	7	66.9	0.1	33
SSP2-4.5	R99p	272	7	30.5	1.1	68.4
SSP2-4.5	Rx1day	273	7	82.3	0.1	17.6
SSP2-4.5	Rx5day	273	7	75.6	0.1	24.4
SSP2-4.5	SDII	273	7	63.6	1.6	34.9
SSP2-4.5	q50	273	7	25.4	5.4	69.2
SSP2-4.5	q90	273	7	45	0.2	54.8
SSP2-4.5	q95	273	7	67.4	0	32.5
SSP2-4.5	q99	273	7	70.6	0	29.4
SSP2-4.5	wet_day_pct	273	7	64	0.2	35.8
SSP5-8.5	CDD	273	7	58.8	0	41.2
SSP5-8.5	CWD	273	7	73.8	0	26.1
SSP5-8.5	PRCPTOT	273	7	83.5	0.2	16.3

Scenario	Index	N cells	GCMs	GCM main effect (%)	BC-method main effect (%)	Residual (%)
SSP5-8.5	PRCPTOT_dry	273	7	77.9	0	22
SSP5-8.5	PRCPTOT_wet	273	7	81.8	0.2	18.1
SSP5-8.5	R10mm	273	7	76.2	0	23.8
SSP5-8.5	R20mm	273	7	42.2	0	57.8
SSP5-8.5	R50mm	214	7	25.4	0	74.5
SSP5-8.5	R95p	273	7	80.2	1	18.9
SSP5-8.5	R99p	272	7	74.5	0.1	25.4
SSP5-8.5	Rx1day	273	7	74.9	0.2	24.8
SSP5-8.5	Rx5day	273	7	74.1	0.2	25.7
SSP5-8.5	SDII	273	7	76.8	0.3	23
SSP5-8.5	q50	273	7	52.5	0.4	47
SSP5-8.5	q90	273	7	72.6	0	27.4
SSP5-8.5	q95	273	7	70.6	0.5	29
SSP5-8.5	q99	273	7	68.6	0.7	30.7
SSP5-8.5	wet_day_pct	273	7	78.2	0	21.8

Supplementary Table S7. Temporal-dependence statistics: mean absolute bias against observations, averaged over gauges and GCMs. Gauge- and model-level values are in Supplementary Data S7.

Period	Method	ACF(1) daily	ACF(1) occurrence	P(wet wet)	P(wet dry)	Mean wet spell	Mean dry spell	p90 dry spell
Calibration	Raw CMIP6	118.14	29.78	30.44	68.48	64.98	46.55	48.24
Calibration	QM/EQM	110.15	32.42	13.98	56.37	30.44	31.11	44.7
Calibration	DetQM	110.15	32.41	13.99	56.18	30.46	31.33	44.97
Calibration	QDM	110.15	32.42	13.98	56.37	30.44	31.11	44.7
Validation	Raw CMIP6	82.77	28.43	26.99	74.76	53.35	54.26	56.28
Validation	QM/EQM	74.73	29.79	14.25	62.55	28.11	35.66	37.37
Validation	DetQM	75.15	29.81	14.32	62.6	28.19	35.97	37.72
Validation	QDM	74.43	29.79	14.4	62.16	28.23	36.43	38.35

Supplementary Table S8. Paired method differences against QM/EQM, model-level and descriptive.

Scenario	Index	DetQM – QM/EQM median (%)	n paired GCMs	QDM – QM/EQM median (%)
SSP2-4.5	Wet days	0.01	7	-0.28
SSP2-4.5	PRCPTOT	-1.11	7	-0.65
SSP2-4.5	SDII	-1.54	7	-0.66
SSP2-4.5	q50	-2.69	7	-0.53
SSP2-4.5	q90	-0.29	7	0.08
SSP2-4.5	q95	-0.2	7	-0.45
SSP2-4.5	q99	-0.25	7	-0.95
SSP2-4.5	Rx1day	-0.41	7	-0.21
SSP2-4.5	R10mm	-1.63	7	-0.99
SSP2-4.5	R20mm	0.49	7	0.08
SSP2-4.5	R50mm	-4.24	6	-2.11
SSP2-4.5	R95p	-1.44	7	-1.04
SSP2-4.5	R99p	0.35	7	-1.22
SSP2-4.5	Rx5day	-0.79	7	-0.98
SSP2-4.5	CDD	-0.06	7	0.75
SSP2-4.5	CWD	0	7	0.21
SSP5-8.5	Wet days	0.01	7	0.06
SSP5-8.5	PRCPTOT	-0.21	7	-0.71
SSP5-8.5	SDII	-0.29	7	-0.34
SSP5-8.5	q50	-1.28	7	-0.48
SSP5-8.5	q90	0.21	7	0.37
SSP5-8.5	q95	-0.36	7	-0.27

Scenario	Index	DetQM – QM/EQM median (%)	n paired GCMs	QDM – QM/EQM median (%)
SSP5-8.5	q99	-0.12	7	-1.28
SSP5-8.5	Rx1day	0.39	7	-1.39
SSP5-8.5	R10mm	0.27	7	-0.72
SSP5-8.5	R20mm	1.39	7	0.12
SSP5-8.5	R50mm	-0.19	6	-3.21
SSP5-8.5	R95p	0.3	7	-1.59
SSP5-8.5	R99p	-2.07	7	3.11
SSP5-8.5	Rx5day	-0.28	7	-0.89
SSP5-8.5	CDD	0.03	7	0.22
SSP5-8.5	CWD	0	7	0.05

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33 **Supplementary Table S9.** Acceptance-gate status.

gate	status	criterion	evidence	limitation
A: observed data	PASS WITH RESIDUAL BIAS	suspicious zeros classified by automated screening; no imputation	classes {'not_zero': 4866, 'confirmed_zero': 315, 'uncertain': 80, 'probable_missing': 43}; 0 values imputed	station metadata not yet verified; 4 gauge(s) inhomogeneous between calibration and validation (351003, 351005, 351006, 351007). QC is closed for automated screening but NOT for station homogenisation.
B: historical bias correction	PASS (MARGINAL ONLY)	calibration-period MARGINAL distribution reproduced ($ \text{bias} < 5\%$); this criterion covers the marginal distribution only and makes no claim about temporal or multi-day structure (see Gate G)	marginal $ \text{bias} $ 0.19%; spell/multi-day $ \text{bias} $ 27.59%	quantile mapping is a marginal transformation and does not reorder days, so CDD, CWD and Rx5day retain the model's own temporal structure. Do not claim that extremes as a class were successfully reproduced.
C: independent validation	PASS WITH RESIDUAL BIAS	out-of-sample $ \text{bias} $ smaller than raw GCM on the primary metrics	improved on 4/4; mean residual $ \text{bias} $ 26.4%; excluding the gauges flagged as non-homogeneous, the network-mean residual is 13.4%	residual out-of-sample error remains substantial. 'Better than raw' does not mean 'reliable for all extremes'; station-level errors are larger than the network mean. The reduction obtained by excluding non-homogeneous gauges is an association between station heterogeneity and validation error, not a causal attribution of a share of that error.
D: future signal	PASS	change = BC_future / BC_historical, same model and method	enforced in code; observations never used as denominator	
E: method comparison	PASS	raw, QM, DetQM and QDM evaluated on an identical split	methods present: ['detqm', 'qdm', 'qm', 'raw']	
F: maps and boundary	PASS	verified administrative boundary, CRS and coordinates	verified	
G: temporal dependence	DIAGNOSTIC PASS - TEMPORAL STRUCTURE NOT CORRECTED (NOT A PREDICTION-PERFORMANCE PASS)	DIAGNOSTIC ONLY - this gate measures and reports temporal dependence; it is not an acceptance criterion for prediction performance. No hard threshold is imposed because QM/DetQM/QDM are marginal	improved vs raw on 3/4 sequencing statistics (P_wet_given_wet, mean_dry_spell, mean_wet_spell); worst residual $ \text{bias} $ 33.77%; not improved: acf1_occurrence	the bias correction improved selected occurrence-transition statistics relative to the raw GCM, but substantial residual biases remain in spell-duration and multi-day statistics. Do not interpret CDD, CWD or Rx5day

gate	status	critereon	evidence	limitation
		transformations and do not explicitly reconstruct temporal sequencing		projections as corrected.
H: raw-data anomaly review	PASS	raw model change reviewed against the regional literature range; flagged cases are quarantined from interpretation, not failed	no anomaly source trace: B3 NetCDF metadata=NOT ESTABLISHED; B4 unit conversion=ESTABLISHED (indirect); B5 calendar=ESTABLISHED; B6 concatenation=ESTABLISHED; B7 grid/station extraction=PARTIAL; B8 raw-only signal=ESTABLISHED	

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35 **Supplementary Table S10.** CMIP6 model inventory.

model	scenario	variant	grid	file	calendar	days_per_year	first	last	n_days	mean_daily_mm	implied_annual_mm	verdict
ACCE-SS-ESM1-5	historical	r1i1p1f1	gn	pr_day_ACCESS-ESM1-5_historical_r1i1p1f1_gn_19810101-20141231_63sta.csv	standard (Gregorian)	[np.int64(365), np.int64(366)]	1981-01-01	2014-12-31	12418	1.5009	548.2	OK: consistent with mm/day
ACCE-SS-ESM1-5	ssp245	r1i1p1f1	gn	pr_day_ACCESS-ESM1-5_ssp245_r1i1p1f1_gn_20150101-21001231_63sta.csv	standard (Gregorian)	[np.int64(365), np.int64(366)]	2015-01-01	2100-12-31	3141	1.5228	556.2	OK: consistent with mm/day
ACCE-SS-ESM1-5	ssp585	r1i1p1f1	gn	pr_day_ACCESS-ESM1-5_ssp585_r1i1p1f1_gn_20150101-21001231_63sta.csv	standard (Gregorian)	[np.int64(365), np.int64(366)]	2015-01-01	2100-12-31	3141	1.226	447.8	OK: consistent with mm/day
CESM2	historical	r11i1p1f1	gn	pr_day_CESM2_historical_r11i1p1f1_gn_19810101-20141231_63sta.csv	365_day (noleap)	[np.int64(365)]	1981-01-01	2014-12-31	12410	2.8099	1026.3	OK: consistent with mm/day
CESM2	ssp245	r11i1p1f1	gn	pr_day_CESM2_ssp245_r11i1p1f1_gn_20150101-21001231_63sta.csv	365_day (noleap)	[np.int64(365)]	2015-01-01	2100-12-31	31390	2.9056	1061.3	OK: consistent with mm/day
CESM2	ssp585	r11i1p1f1	gn	pr_day_CESM2_ssp585_r11i1p1f1_gn_20150101-21001231_63sta.csv	365_day (noleap)	[np.int64(365)]	2015-01-01	2100-12-31	31390	2.5124	917.6	OK: consistent with mm/day
CanESM5	historical	r1i1p1f1	gn	pr_day_CanESM5_historical_r1i1p1f1_gn_19810101-20141231_63sta.csv	365_day (noleap)	[np.int64(365)]	1981-01-01	2014-12-31	12410	4.2149	1539.5	OK: consistent with mm/day
CanESM5	ssp245	r1i1p1f1	gn	pr_day_CanESM5_ssp245_r1i1p1f1_gn_20150101-21001231_63sta.csv	365_day (noleap)	[np.int64(365)]	2015-01-01	2100-12-31	31390	4.0529	1480.3	OK: consistent with mm/day
CanESM5	ssp585	r1i1p1f1	gn	pr_day_CanESM5_ssp585_r1i1p1f1_gn_20150101-21001231_63sta.csv	365_day (noleap)	[np.int64(365)]	2015-01-01	2100-12-31	31390	3.3705	1231.1	OK: consistent with mm/day
EC-Earth3	historical	r1i1p1f1	gr	pr_day_EC-Earth3_historical_r1i1p1f1_gn_19810101-20141231_63sta.csv	standard (Gregorian)	[np.int64(365), np.int64(366)]	1981-01-01	2014-12-31	12418	1.3686	499.9	OK: consistent with mm/day
EC-Earth3	ssp245	r1i1p1f1	gr	pr_day_EC-	standard	[np.int64(365), np.int64(366)]	2015-01-01	2100-12-31	3141	1.5231	556.3	OK: consistent with mm/day

model	scena rio	vari ant	g ri d	file	calendar	days_per_year	first	last	n_ da ys	mean_ daily_ mm	implied_ annual_ mm	verdict
Earth3	5	1f1		Earth3_ssp245_r1i1p1f1_gr_20150101-21001231_63sta.csv	(Gregorian)	np.int64(366)]	01-01	12-31	41 1			with mm/day
EC-Earth3	ssp585	r1i1p1f1	gr	pr_day_EC-Earth3_ssp585_r1i1p1f1_gr_20150101-21001231_63sta.csv	standard (Gregorian)	[np.int64(365), np.int64(366)]	2015-01-01	2100-12-31	31 41 1	1.4538	531	OK: consistent with mm/day
FGOA-LS-g3	historical	r1i1p1f1	g	pr_day_FGOALS-g3_historical_r1i1p1f1_gn_19810101-20141231_63sta.csv	365_day (noleap)	[np.int64(365)]	1981-01-01	2014-12-31	12 41 0	0.7866	287.3	OK: consistent with mm/day
FGOA-LS-g3	ssp245	r1i1p1f1	g	pr_day_FGOALS-g3_ssp245_r1i1p1f1_gn_20150101-21001231_63sta.csv	365_day (noleap)	[np.int64(365)]	2015-01-01	2100-12-31	31 39 0	0.6907	252.3	OK: consistent with mm/day
FGOA-LS-g3	ssp585	r1i1p1f1	g	pr_day_FGOALS-g3_ssp585_r1i1p1f1_gn_20150101-21001231_63sta.csv	365_day (noleap)	[np.int64(365)]	2015-01-01	2100-12-31	31 39 0	0.6254	228.4	OK: consistent with mm/day
MIRO-C6	historical	r1i1p1f1	g	pr_day_MIROC6_historical_r1i1p1f1_gn_19810101-20141231_63sta.csv	standard (Gregorian)	[np.int64(365), np.int64(366)]	1981-01-01	2014-12-31	12 41 8	3.0912	1129.1	OK: consistent with mm/day
MIRO-C6	ssp245	r1i1p1f1	g	pr_day_MIROC6_ssp245_r1i1p1f1_gn_20150101-21001231_63sta.csv	standard (Gregorian)	[np.int64(365), np.int64(366)]	2015-01-01	2100-12-31	31 41 1	3.1708	1158.1	OK: consistent with mm/day
MIRO-C6	ssp585	r1i1p1f1	g	pr_day_MIROC6_ssp585_r1i1p1f1_gn_20150101-21001231_63sta.csv	standard (Gregorian)	[np.int64(365), np.int64(366)]	2015-01-01	2100-12-31	31 41 1	3.046	1112.6	OK: consistent with mm/day
MRI-ESM2-0	historical	r1i1p1f1	g	pr_day_MRI-ESM2-0_historical_r1i1p1f1_gn_19810101-20141231_63sta.csv	standard (Gregorian)	[np.int64(365), np.int64(366)]	1981-01-01	2014-12-31	12 41 8	2.4819	906.5	OK: consistent with mm/day
MRI-ESM2-0	ssp245	r1i1p1f1	g	pr_day_MRI-ESM2-0_ssp245_r1i1p1f1_gn_20150101-21001231_63sta.csv	standard (Gregorian)	[np.int64(365), np.int64(366)]	2015-01-01	2100-12-31	31 41 1	2.2751	831	OK: consistent with mm/day
MRI-ESM2-0	ssp585	r1i1p1f1	g	pr_day_MRI-ESM2-0_ssp585_r1i1p1f1_gn_20150101-21001231_63sta.csv	standard (Gregorian)	[np.int64(365), np.int64(366)]	2015-01-01	2100-12-31	31 41 1	1.8167	663.5	OK: consistent with mm/day

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37 Supplementary Data files

38 The complete listings behind the aggregated tables above are provided as comma-separated files in this
39 submission.

40

File	Rows	Columns
Supplementary_Data_S2.csv	364	67
Supplementary_Data_S3.csv	364	67
Supplementary_Data_S5.csv	9804	11
Supplementary_Data_S7.csv	728	43
Supplementary_Data_S10b_fit_diagnostics.csv	91	9
Supplementary_Data_S10c_apply_diagnostics.csv	546	10